

NE555 Circuit Schematic Vision Analysis Report

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INTERNAL

Source Document:	circuit_diagram_e2e.png	Analysis Task:	VISION_ANALYSIS
AI Model:	qwen3-vl:4b	Sovereignty Mode:	SOVEREIGN ON-PREMISE (AIR-GAPPED)

EXECUTIVE SUMMARY

Visual schematic inspection of circuit_diagram_e2e.png reveals a standard 555 timer pulse generator circuit.

KEY FINDINGS

- IC1 NE555 timer configured for astable oscillation
- Timing network: R1=10k, R2=47k, C1=10uF
- Output LED indicator driven via pin 3

DETAILED ANALYSIS

The circuit implements standard timer oscillation. Bypass capacitor C2 stabilizes control voltage on pin 5.

RISKS AND OPERATIONAL ISSUES

Operating supply voltage must not exceed 15V DC to prevent thermal breakdown of IC1.

RECOMMENDATIONS

Ensure polarized capacitor C1 is installed with correct polarity orientation.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
circuit_diagram_e2e.png	Page(s) 1	Verified Visual Artifact

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